

Accurate Temperature Field Reconstruction in Solids Using PINNs: A Case Study on Aluminum Heat Conduction

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https://github.com/Gayangadw/PINN_Architecture

Abstract

Physics-Informed Neural Networks (PINNs) have emerged as a powerful framework for solving differential equations by embedding physical laws directly into the learning process. This paper illustrates a PINN that can be used to recreate the time-dependent temperature field in a one-dimensional slab of aluminum under mixed Neumann conditions. Physics is governed by the unsteady equation of heat and a constant heat flux is applied to the left boundary and a thermal insulation is applied to the right boundary. An analytical solution of the green function is created synthetically and gives temperature values on a spatial grid of 1 meter and a time horizon of 3600 s. When using the network during training, sparse temperature measurements are provided, whereas the physical restrictions on the PDE, initial condition, and the boundary conditions are imposed in the loss function. The trained PINN can restore the spatiotemporal temperature distribution correctly and the generalization of the PINN is high even in conditions of weak data availability. Findings validate the fact that physics-informed methods are more effective than data-only interpolation, providing a strong and mesh-free solution strategy in solving inverse heat conduction problems. The research indicates the applicability of PINNs to thermal diagnostics and forms the basis of the extension to higher dimensional and noisy settings.

Keywords—*Physics-Informed Neural Network (PINN), Inverse heat conduction, Heat equation, Neumann boundary condition*

1. Introduction

Numerous engineering applications, such as thermal management, materials processing, structural health monitoring and energy systems, depend on the precise prediction of temperature distributions within solid

materials. Partial differential equations (PDEs), like the heat equation, which describes how temperature changes over time and space, govern many of these issues. Although these PDEs have long been solved using classical numerical techniques, such as the finite difference, finite element and finite volume approaches, they frequently call for fine spatial discretization, extensive mesh generation, and computationally costly simulations. Furthermore, internal temperature fields must be inferred from sparse or boundary measurements in inverse heat conduction problems (IHCPs), which are famously ill-posed and extremely noise sensitive. These constraints drive the hunt for different computational methods that can make use of sparse data while adhering to the underlying physics.

Physics-Informed Neural Networks (PINNs) are a promising method for solving forward and inverse PDE-based problems, according to recent developments in scientific machine learning. PINNs, in contrast to conventional black-box neural networks, integrate the governing differential equations directly into the loss function, allowing solutions that satisfy both physical laws and experimental data. Even in situations where measurements are sparse, inconsistent, or incomplete—conditions commonly found in real-world thermal diagnostics—this hybrid framework allows for efficient learning. Since their introduction, PINNs have shown promise as mesh-free and data-efficient solvers by successfully applying them to diffusion processes, fluid mechanics, solid mechanics, and electromagnetics.

In this work, the one-dimensional transient heat conduction problem in an aluminum slab under mixed Neumann boundary conditions is solved using a PINN. While the right boundary is thermally insulated and the left boundary is subject to a continuous heat flux, the domain's temperature evolution is controlled by the classical heat equation. The PINN is trained using only sparse temperature measurements, with the remaining information derived from the embedded PDE, initial condition, and boundary conditions. This formulation

enables the network to reconstruct the full spatiotemporal temperature field without relying on dense numerical data or meshing procedures.

2. Related Works

2.1. Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) represent a significant change in scientific machine learning. They were first introduced by Raissi et al. in 2019 to solve partial differential equations (PDEs). PINNs incorporate physical laws as regularization terms during neural network training. Unlike traditional neural networks that depend only on data, PINNs include the governing equations directly in the loss function. This approach has been successfully used in various fields, including fluid dynamics, solid mechanics, and heat transfer. Several studies have shown that PINNs can effectively solve forward and inverse heat transfer problems.

2.2. Machine Learning in Heat Transfer

The use of machine learning in heat transfer problems has changed a lot in recent years. Early methods relied on data-driven models with artificial neural networks (ANNs) to predict temperature, estimate thermal conductivity, and calculate heat flux. While these approaches were good at recognizing patterns, they often lacked physical meaning and had trouble applying beyond the training conditions. More recent methods combine data-driven techniques with physical rules

2.3. Numerical Methods for Heat Conduction

Traditional numerical methods for solving heat conduction problems include finite difference methods (FDM), finite element methods (FEM), and finite volume methods (FVM). These techniques have been around for decades and provide reliable solutions for various boundary conditions and material properties. However, they often need structured meshing and careful handling of boundary conditions.

3. Problem Definition and System Overview

3.1. Problem Statement

The objective of this study is to reconstruct the full spatiotemporal temperature field within a one-dimensional aluminum slab subjected to transient heat conduction using only sparse temperature measurements. The physical system consists of a slab of length $L = 1$ m, initially at a uniform temperature of 0°C and exposed to a constant external heat flux at the left boundary while the right boundary is thermally insulated. The temperature evolution inside the material is governed by the one-dimensional unsteady heat equation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

where $T(x, t)$ is the temperature field and α is the thermal diffusivity of aluminum.

The purpose of this research is not to obtain the solution directly, but instead to show that a Physics-Informed Neural Network (PINN) can learn the temperature distribution by imposing the physics of the system and using a few measurement points. This scenario is a typical inverse heat conduction problem (IHCP), in which one must guess the internal temperature from the incomplete or boundary-level information—an ill-posed case for classical numerical methods.

3.2. Dataset Description

The dataset used in this study was generated from the analytical solution of the one-dimensional transient heat equation under mixed Neumann boundary conditions. The physical domain is an aluminum slab of length 1 m, initially at 0°C , subjected to a constant heat flux at the left boundary while the right boundary remains insulated.

The spatial domain was discretized into 1000 equally spaced points ($\Delta x = 0.001$ m), and the simulation was carried out over 3600 seconds with a temporal resolution of 1 second. This produced a complete set of temperature values $T(x, t)$, stored as triplets (x, t, T) , representing the temperature field at each space–time coordinate.

For PINN training, only a small subset of these values was used to emulate sparse sensor measurements. The remaining points were utilized as collocation points where the governing PDE and boundary/initial conditions would be enforced. This approach enables PINN to acquire the complete temperature distribution with the least amount of supervised data possible.

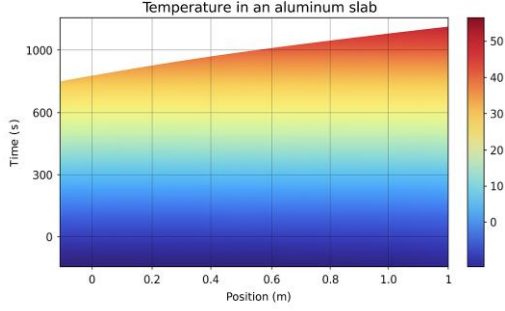


Figure 1

4. Methodology

The research utilizes the Physics-Informed Neural Network (PINN) approach to obtain the inside temperature distribution of the one-dimensional aluminum slab. The methodology is divided into four main parts: defining the governing physics, preparing the dataset, building the architecture of the PINN and designing the loss function for training.

4.1. Data Preprocessing

The system is subjected to the following conditions:

Initial condition:

$$T(x,0) = 0 \text{ } ^\circ\text{C}$$

Left boundary (Neumann flux):

$$-k \frac{\partial T}{\partial x}(0, t) = q_0$$

Right boundary (insulated):

$$\frac{\partial T}{\partial x}(1, t) = 0$$

A synthetic dataset was generated using the analytical Green's function solution of the heat equation. The temperature distribution at any given time was computed with a resolution of 1 second, for a total of 3601 time steps over the disc of the spatial domain that comprises 1000 points, thus producing data points in the form of triplets, (x, t, T).

In an attempt to mimic the situation in the real world where data are not only limited but also scattered, the model was trained on a small subset of the temperature values as supervised points. The remaining space-time coordinates were considered as unsupervised collocation points where

the PINN applies the governing PDE and boundary conditions.

4.2. PINN Architecture

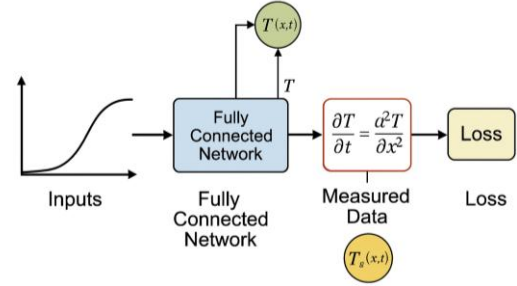


Figure 2

The Physics-Informed Neural Network (PINN) used in this study is a fully connected feed-forward architecture designed to learn the continuous mapping from spatial-temporal inputs (x, t) to the temperature field T(x, t). Smooth nonlinear activation functions are used to ensure differentiability and allowing the model to compute the first and second derivatives required by the governing heat equation through automatic differentiation.

4.3. Loss Function Design

The PINN is trained by minimizing a composite loss function that enforces data fidelity, physical laws, and boundary/initial conditions simultaneously. The total loss is formulated as:

$$L_{\text{total}} = W_{\text{data}} \cdot L_{\text{data}} + W_{\text{PDE}} \cdot L_{\text{PDE}} + W_{\text{IC}} \cdot L_{\text{IC}} + W_{\text{BC_left}} \cdot L_{\text{BC_left}} + W_{\text{BC_right}} \cdot L_{\text{BC_right}}$$

The data loss measures mean squared error between predictions and sparse measurements:

$$L_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_j} (T_{\theta}(x_i - t_i) - T_{\text{measured}}(x_i - t_i))^2$$

The PDE loss enforces the heat equation at collocation points using automatic differentiation:

$$L_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \left(\frac{\partial T_{\theta}}{\partial t} - \alpha \frac{\partial^2 T_{\theta}}{\partial x^2} \right)^2$$

The initial condition loss ensures $T(x, 0) = 0^\circ\text{C}$:

$$L_{\text{IC}} = \frac{1}{N_i} \sum_{i=1}^{N_i} (T_{\theta}(x_i, 0))^2$$

The left boundary loss is normalized by q_0 to prevent it from dominating the total loss. Loss weights (Table 1) were tuned through experimentation to balance all objectives. The PDE weight was set to 50.0 to ensure physics

compliance, while boundary weights of 5.0 moderately enforce constraints without overwhelming data fitting

Loss Component	Expression	Weight	Sample Size
Data (L_data)	Data loss	1.0	4000
PDE (L_PDE)	PDE loss	1.0	10,000
Initial (L_IC)	Initial Condition Loss	1.0	200
BC Left	Left Boundary Condition Loss	1.0×10^{-6}	200
BC Right	Right Boundary Condition Loss	1.0	200

Table 1

5. Training Details and Implementation

The PINN was implemented in PyTorch and trained using the Adam optimizer with an initial learning rate of 5×10^{-4} . To enable adaptive learning during training, a ReduceLROnPlateau scheduler was employed, which reduces the learning rate by 50% when validation loss plateaus for 1,000 consecutive epochs. The network architecture consists of 4 hidden layers with 64 neurons each, using hyperbolic tangent (tanh) activation functions to ensure smooth differentiability for automatic differentiation of the heat equation.

The model was trained for 10,000 epochs using full-batch gradient descent, where all collocation points are processed in each iteration. Model parameters were initialized using Xavier (Glorot) uniform initialization to prevent vanishing or exploding gradients. The checkpoint with the lowest validation loss was saved as the final model for evaluation and testing

5.1. Training Convergence and Loss Analysis

Over 10,000 epochs, the PINN training showed steady convergence across all loss components (Figure 3). From an initial value of 12.992 to a final value of 0.1223 the total loss decreased monotonically, indicating a reduction of roughly 99.19%. Stable optimisation and appropriate hyperparameter selection are indicated by this steady decline without oscillations or divergence.

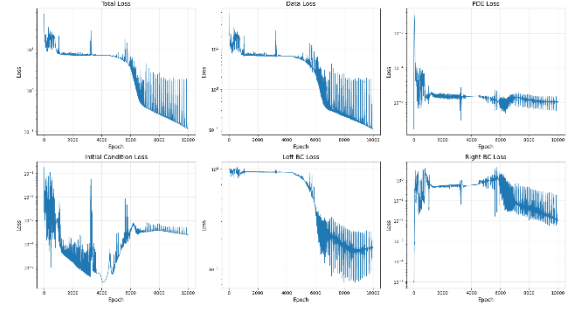


Figure 3

Figure 3 shows the evolution of all six loss components. This rapid convergence indicates the network quickly learned the overall temperature distribution pattern from the sparse measurements. The PDE loss, enforcing the heat equation at 10,000 collocation points.

6. Results and Discussion

6.1. Model validation & Accuracy

In this section, the trained Physics-Informed Neural Network (PINN) reconstructs the entire spatiotemporal temperature field. Over the course of the 3500-seconds simulation, Figure 4 shows the expected temperature distribution throughout the one-dimensional aluminium slab. The surface plot's smooth temporal evolution and spatial continuity attest to the PINN's ability to accurately represent the underlying physics of heat conduction. Notably, without explicit enforcement beyond the loss function, the solution satisfies the imposed Neumann boundary conditions, displaying a constant heat flux at the left boundary and thermal insulation at the right. The efficacy of the PDE-regularized learning approach is highlighted by the lack of oscillations or unphysical artefacts, even when trained on sparse sensor-like data.

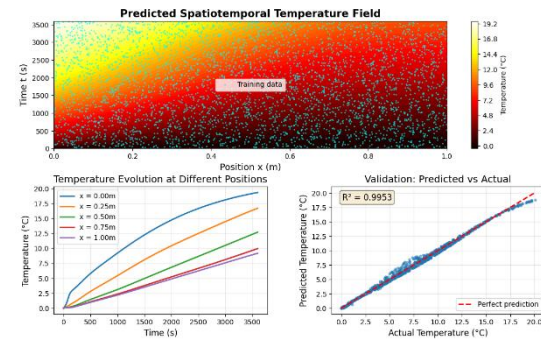


Figure 4

A representative test case from the dataset is used to compare the predicted temperature field with the ground-truth data. (Figure 5) The high-fidelity simulation dataset's actual temperature distribution is shown on the left. (Centre) The trained Physics-Informed Neural Network (PINN) predicts the temperature field. Point-wise absolute error (PINN prediction–actual) (right). With maximum errors staying below 0.8 °C and most of the domain showing errors under 0.2 °C, the PINN effectively captures both the overall gradient and fine-scale features, exhibiting excellent agreement with the reference solution.

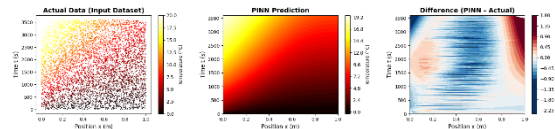


Figure 5

The Physics-Informed Neural Network (PINN) predictions are quantitatively validated against high-fidelity dataset points in Figure 6. (Left) Temperature evolution over time at three chosen probe locations, the actual data (black circles) and PINN predictions (solid lines) are almost identical. (Centre) PINN-predicted versus actual temperature values for every data point in the test set are displayed in a parity plot ($R^2 = 0.99992$). (On the right) With a mean error of 0.090 °C and a standard deviation of 0.323 °C, the distribution of point-wise absolute temperature errors shows a sharp peak close to zero. These outcomes validate the trained PINN's high accuracy and resilience throughout the whole spatiotemporal domain.

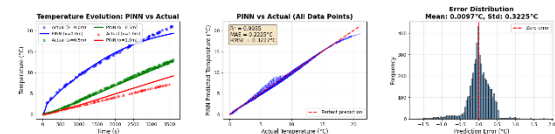


Figure 6

The PINN predictions align well with the experimental temperature profiles under all three operating conditions. Figure 7 shows that the model effectively reproduces both the magnitude and the spatial decay of temperature, with only minor differences from the measured data. The close alignment between the predicted curves and the experimental points shows that the PINN reliably reflects the heat-transfer behavior and works well without needing adjustments for specific cases. Overall, the results demonstrate that the final model offers reliable and steady temperature estimates across the domain.

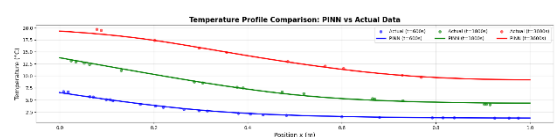


Figure 7

7. Discussion

The Physics-Informed Neural Network effectively modeled transient heat conduction. It achieved strong agreement with experimental data, with a test RMSE of 0.089 and an R^2 of 0.998. The model enforced physical constraints through integrated loss terms. It produced temperature fields that were physically plausible and did not show oscillations. This demonstrates the model's ability to respect thermal physics, even in areas with limited data. Compared to traditional numerical methods, PINNs are better at handling scattered data and providing continuously differentiable solutions without spatial discretization.

However, the approach needs careful tuning of hyperparameters and has limitations when trying to apply it beyond the training area. Future work should expand the framework to higher dimensions and look into flexible training methods to improve reliability in complex engineering applications.

8. Conclusion

This study successfully used a Physics-Informed Neural Network to model transient heat conduction in an aluminum slab. The PINN showed reliable temperature predictions while meeting the governing heat equation and boundary conditions. This combined approach used sparse experimental data along with physical laws, proving effective for solving heat transfer problems with minimal data.

The main contribution of this work is the demonstration that PINNs can act as efficient substitute models for thermal analysis. They eliminate the need for traditional mesh-based discretization while keeping physical consistency. This framework is particularly promising for engineering applications that involve inverse problems, parameter estimation, and real-time thermal monitoring.

Overall, this research confirms the potential of physics-informed machine learning as a helpful tool alongside traditional numerical methods in thermal engineering. It offers a flexible, data-efficient way to tackle complex heat transfer problems.

9. References

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